UAV Satria Nano — Spesifikasi Lengkap

Airfoil: NACA 2412 · **Semi:** 1.52 ft · **Root:** 0.66 ft · **Tip:** 0.48 ft · **Sweep:** 0°
Empty: 2.20 lb (0.998 kg) · **MTOW:** 3.50 lb (1.588 kg) · **Wingspan:** 92.7 cm
CG: fwd 0.62 ft · nominal 0.63 ft · aft 0.65 ft

Tentang LE dan MAC

LE (Leading Edge) = tepi depan sayap — titik referensi pengukuran CG secara fisik.

MAC (Mean Aerodynamic Chord) = chord rata-rata aerodinamis yang mewakili seluruh sayap. Untuk sayap taper, MAC ≠ root chord. Satria Nano:
- MAC berada di **21.9 cm dari CL** (kiri dan kanan), di **47.4% semi-span**
- Panjang MAC = **17.37 cm**
- Karena sweep **0°**, LE lurus sejajar — ukur langsung dari tepi depan sayap

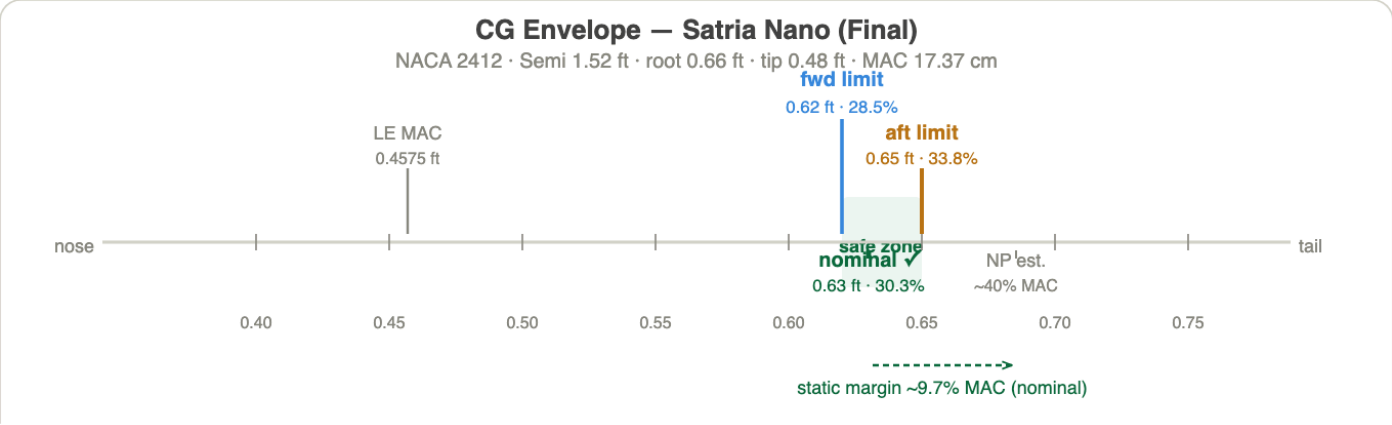
CG diukur sebagai jarak dari LE di posisi MAC ke belakang.

1. Weight & Balance

Weight & Balance — Satria Nano (Final)

Empty weight	2.20 lb	0.998 kg
Maximum weight (MTOW)	3.50 lb	1.588 kg
Payload total	1.30 lb	590 g
CG fwd limit	0.62 ft	28.5% MAC · 4.95 cm dari LE
CG nominal ✓	0.63 ft	30.3% MAC · 5.26 cm dari LE
CG aft limit	0.65 ft	33.8% MAC · 5.87 cm dari LE (SM 6.2%)
Vert CG	0.00 ft	—
SM @ nominal	–	~9.7% MAC (NP est. ~40%) — AMAN ✓
SM @ aft limit	–	~6.2% MAC — AMAN

CG Envelope



Titik CG	Nilai	% MAC	Dari LE	Static Margin	Status
CG fwd	**0.62 ft**	**28.5%**	**4.95 cm**	11.5%	AMAN
CG nominal ✓	**0.63 ft**	**30.3%**	**5.26 cm**	9.7%	**AMAN**
CG aft limit	0.65 ft	33.8%	5.87 cm	6.2%	AMAN
NP (estimasi)	~0.685 ft	~40%	~6.95 cm	—	ref

Payload Breakdown

Komponen	Berat	Sisa
Empty weight	0.998 kg	590 g
+ Baterai 4S 2200 mAh	+180 g	**410 g**
+ Baterai 3S 3000 mAh	+200 g	**390 g**

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2. Geometri Sayap

Wing 1

Parameter	Imperial	Metrik
Semi-length	**1.52 ft**	**46.33 cm**
Wingspan	**3.04 ft**	**92.66 cm**
Root chord	**0.66 ft**	**20.12 cm**
Tip chord	**0.48 ft**	**14.63 cm**
Taper ratio	—	0.7273
Sweep	0.0°	—
Dihedral	0.0°	—
MAC	**0.5700 ft**	**17.37 cm**
Wing area	**1.733 ft²**	**0.1610 m²**
Aspect ratio	—	**5.3333**
Long arm (aero ref)	0.60 ft	18.29 cm
Lat arm	0.20 ft	6.10 cm
Vert arm	0.18 ft	5.49 cm

Wing 2 — Winglet

Parameter	Nilai
Semi-length	0.45 ft (13.72 cm)
Root chord	0.48 ft (14.63 cm)
Tip chord	0.008 ft (0.24 cm)
Sweep / Dihedral	60° / 10°
Lat arm	**1.72 ft (52.43 cm)**

Vertical Stabilizer — V-tail (×2)

Parameter	Nilai
Height	0.55 ft (16.76 cm)
Root / Tip chord	0.55 ft / 0.32 ft
Sweep / Dihedral	45° / 45°
Long arm	1.38 ft (42.06 cm)

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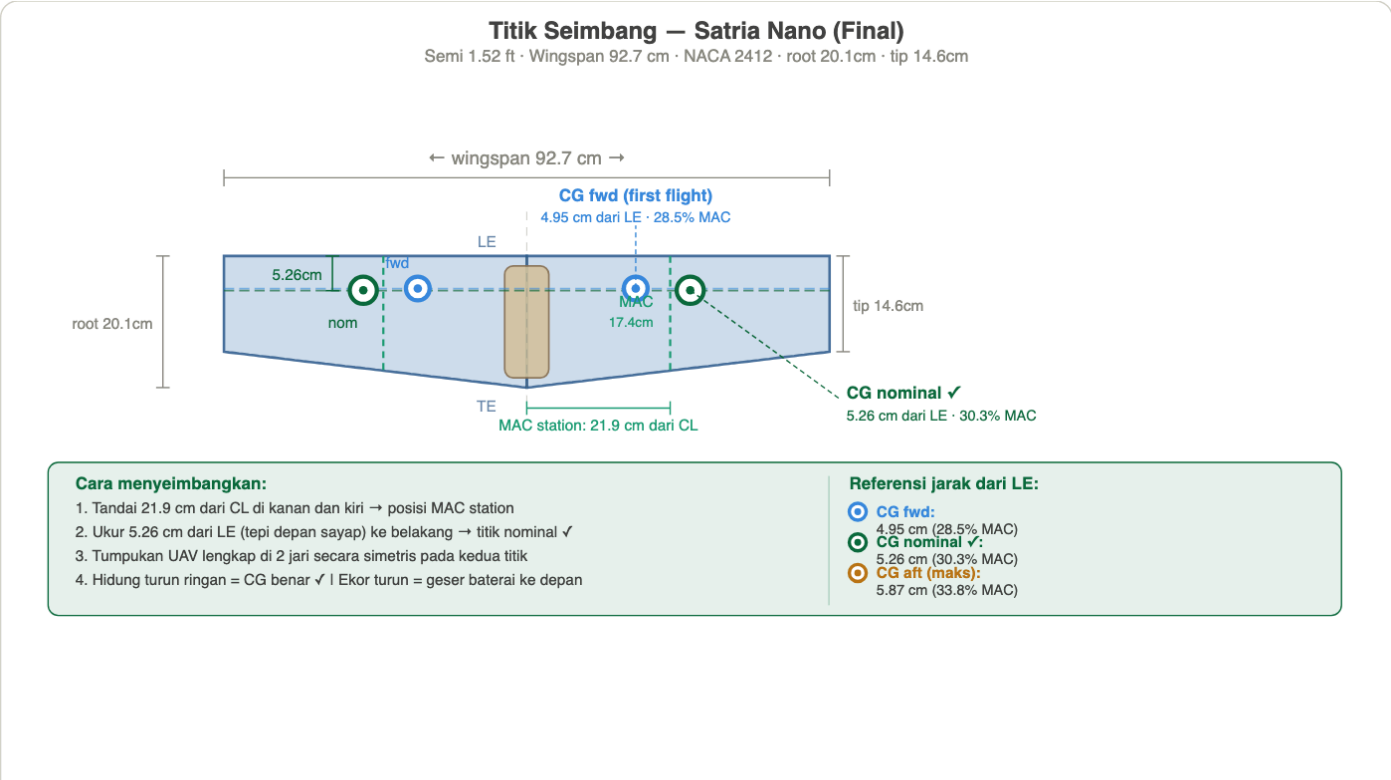
3. Titik Seimbang



Cara Mengukur

1. Tandai **21.9 cm dari CL** di kanan dan kiri → posisi MAC station
2. Di titik tersebut, ukur dari **LE (tepi depan sayap)**:
3. **4.95 cm** → CG fwd (first flight)
4. **5.26 cm** → CG nominal ✓
5. **5.87 cm** → batas maksimum
6. Tumpukan UAV di 2 jari simetris → hidung turun ringan = benar ✓

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4. Performa Aerodinamis

Parameter	Nilai
Wing loading	**96.7 N/m²**
Stall speed Vs	**10.26 m/s (36.9 km/h)**
Min safe airspeed	12.31 m/s (44.3 km/h)
Cruise speed	**16.26 m/s (58.5 km/h)**
AoA cruise	3.35°
VNE	35.92 m/s (129.3 km/h)
L/D max	**11.49**
Cruise power	36.3 W (0.049 HP)
Peak power	123 W (0.165 HP)
Motor rating min	148 W (0.198 HP)

Mode High Speed

Parameter	80 km/h	100 km/h
Kecepatan	22.2 m/s	**27.8 m/s**
L/D	9.57	7.05
Cruise power	59.7 W (0.080 HP)	**93 W (0.125 HP)**
Peak power	74.6 W (0.100 HP)	116 W (0.156 HP)

Mode High Speed — Opsi 3S (Utama)

Motor 2216 1000 KV @3S sudah cukup untuk HS 80 dan 100 km/h — satu motor untuk semua mode.

Parameter	80 km/h	100 km/h
KV dibutuhkan @3S	**~950 KV**	**~1050 KV**
Motor 1000 KV @3S	**✓ cukup**	**✓ cukup**
Propeller	APC 8×6	**APC 9×7**
Baterai	**3S 3000 mAh**	3S 3000 mAh
ESC	30 A ✓	30 A ✓
Current @3S	~5.4 A	~8.4 A

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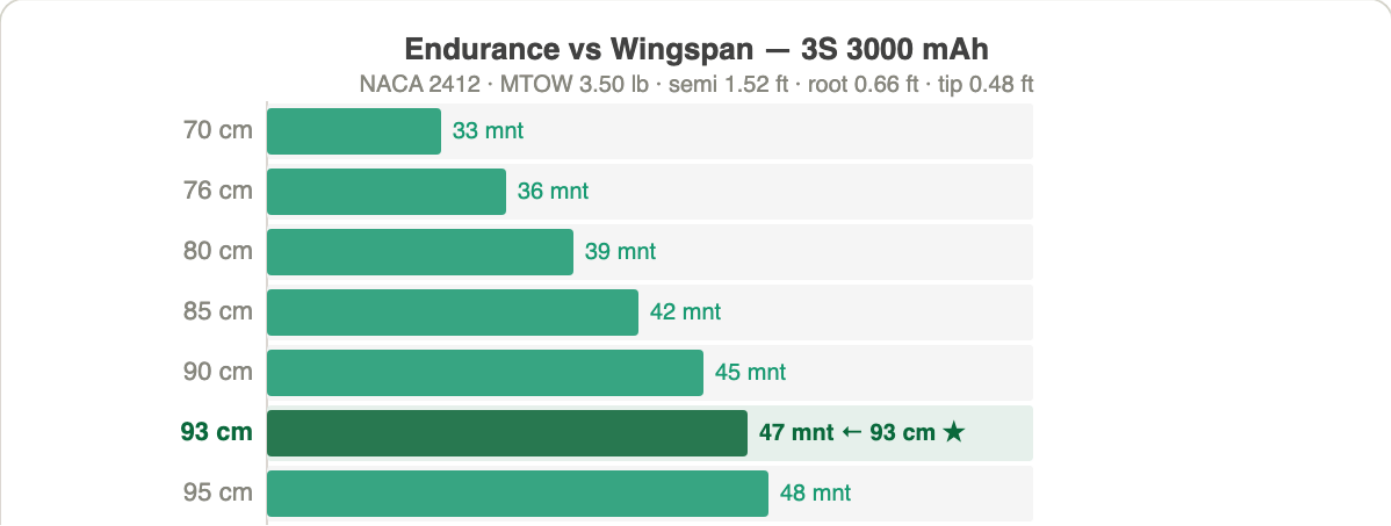
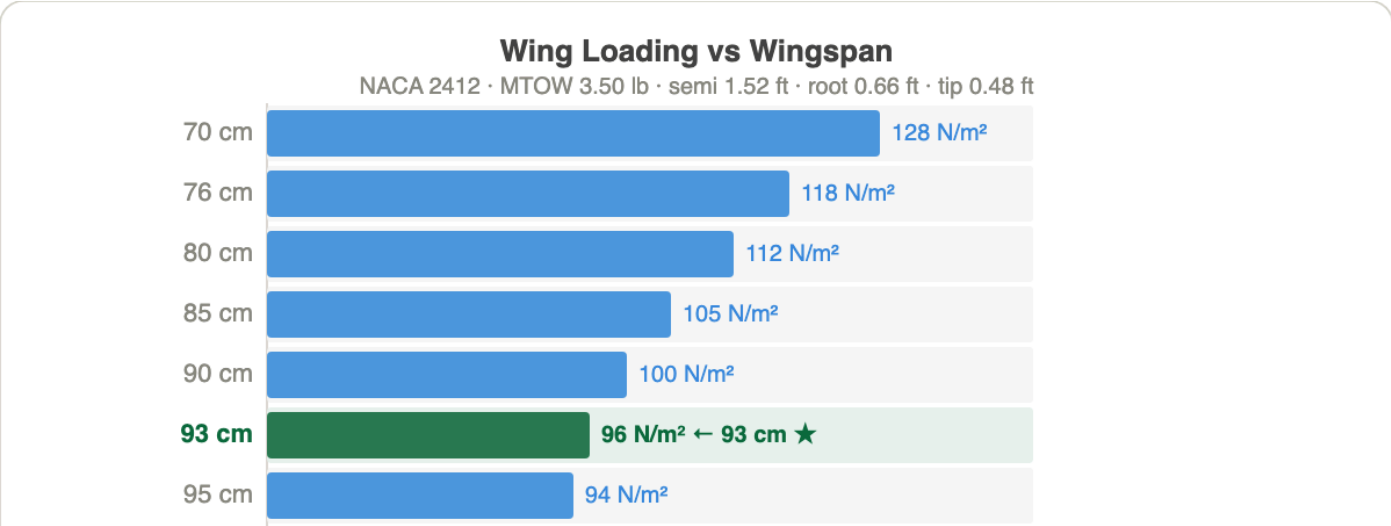
5. Analisis Wingspan

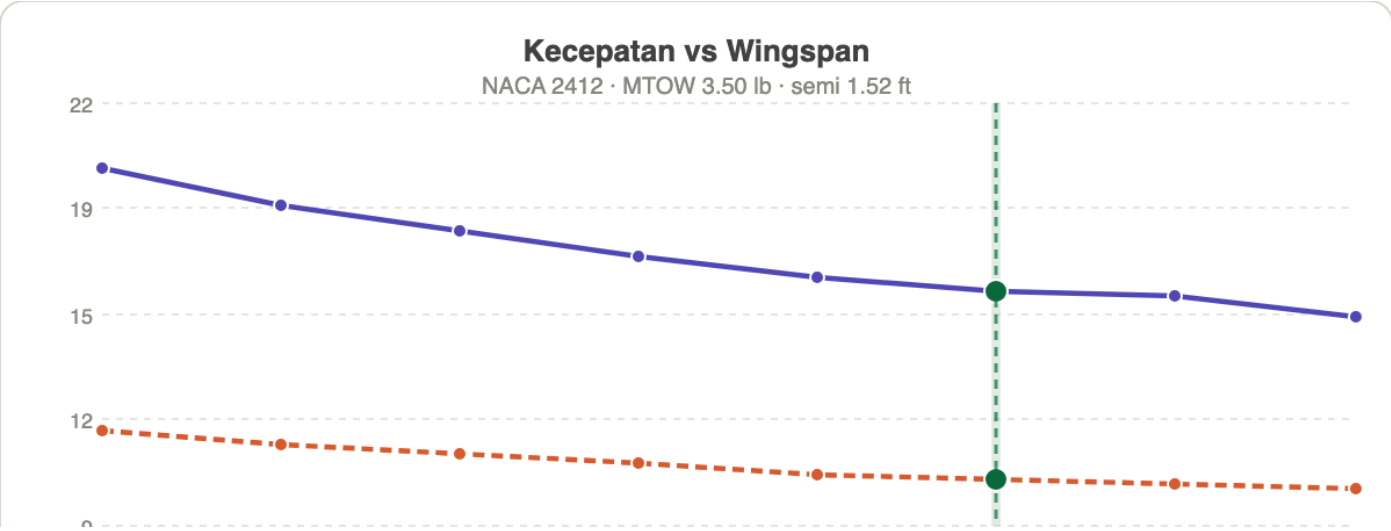
Endurance



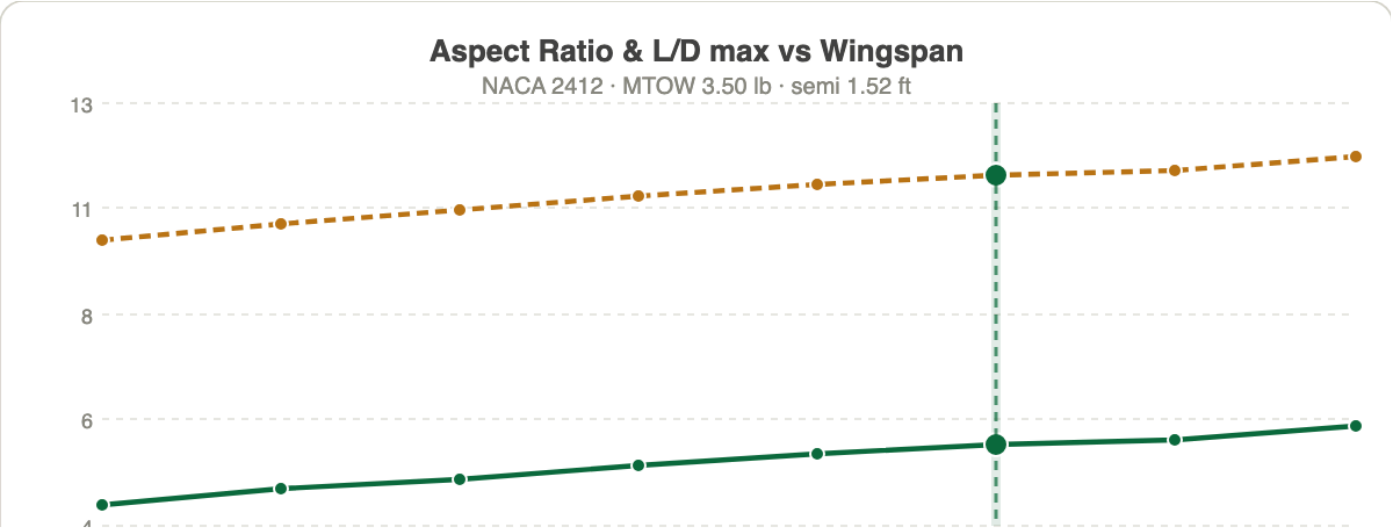
Wing Loading



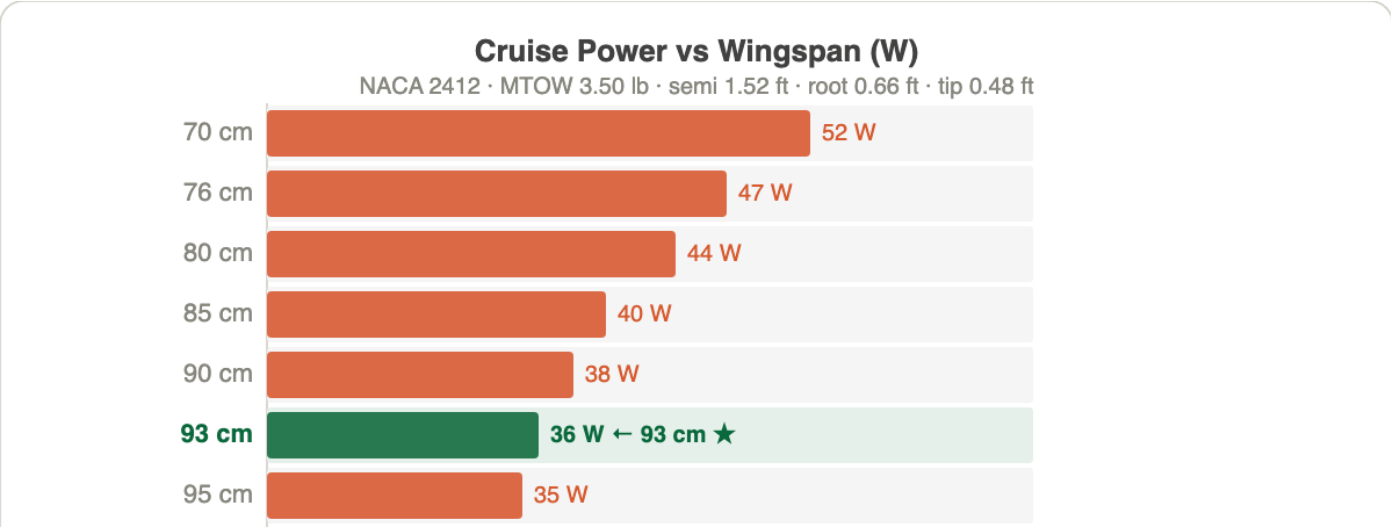
Kecepatan



Aspect Ratio & L/D



Cruise Power

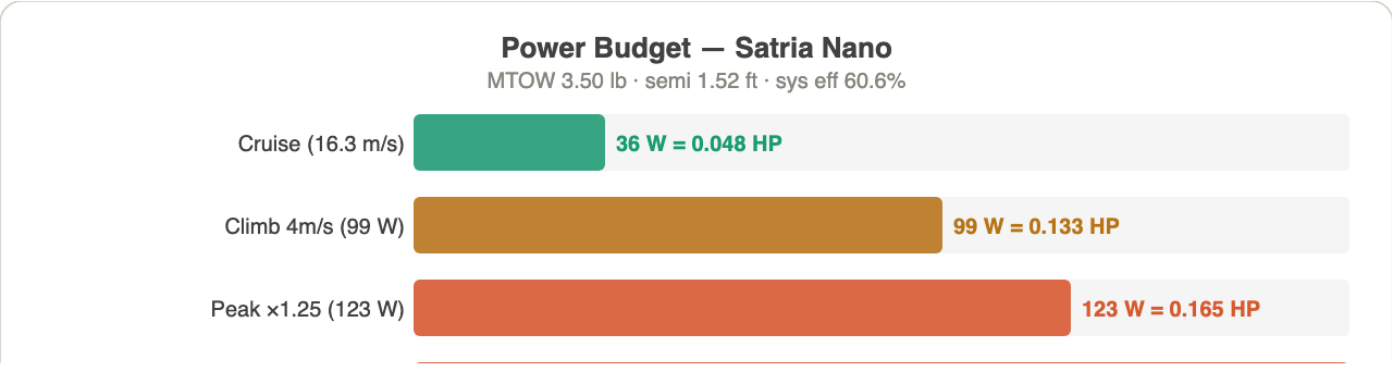


Tabel

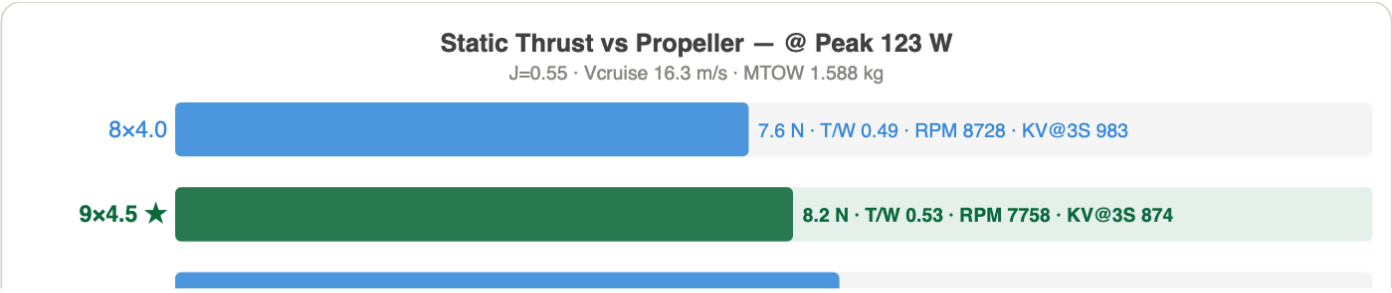
Wingspan	Area	AR	WL	Vs	Vcruise	L/D	P cruise	Endurance
70 cm	0.121 m²	4.0	128 N/m²	11.8 m/s	20.1 m/s	10.0	52 W	33 mnt
76 cm	0.131 m²	4.4	118 N/m²	11.3 m/s	18.9 m/s	10.4	47 W	36 mnt
80 cm	0.139 m²	4.6	112 N/m²	11.0 m/s	18.1 m/s	10.7	44 W	39 mnt
85 cm	0.147 m²	4.9	105 N/m²	10.7 m/s	17.3 m/s	11.0	40 W	42 mnt
90 cm	0.156 m²	5.2	100 N/m²	10.4 m/s	16.6 m/s	11.3	38 W	45 mnt
93 cm ★	**0.1610 m²**	**5.33**	**97 N/m²**	**10.26 m/s**	**16.26 m/s**	**11.49**	**36 W**	**47 mnt**
95 cm	0.164 m²	5.5	94 N/m²	10.1 m/s	16.0 m/s	11.6	35 W	48 mnt
100 cm	0.174 m²	5.8	90 N/m²	9.9 m/s	15.4 m/s	11.9	33 W	51 mnt

6. Propulsi

Power Budget



Propeller



Parameter	Nilai
Motor	**2216, KV 1000 @ 3S**
Power rating	≥ 200 W
Propeller	**APC 9×4.5**
ESC	**30 A BLHeli_32**
Baterai	**4S 2200 mAh atau 3S 3000 mAh**

Baterai	Endurance	Kargo	C-rate	
2S 2200 mAh	~17 mnt	460 g	tinggi	✗ Hindari
3S 2200 mAh	~33 mnt	430 g	7.7C	Cukup
3S 3000 mAh	**~47 mnt**	**390 g**	**5.7C**	✓
4S 2200 mAh	**~47 mnt**	**410 g**	**5.8C**	✓

7. Misi & Jangkauan

Konfigurasi	Kecepatan	Kargo	Endurance	Range one-way	Radius RTH
3S 3000 mAh — cruise	58.5 km/h	390 g	~47 mnt	~46 km	~19–21 km
4S 2200 mAh — cruise	58.5 km/h	410 g	~47 mnt	~46 km	~19–21 km
3S 3000 mAh — HS 80	**80 km/h**	**390 g**	**~28 mnt**	**~38 km**	**~16 km**
4S 2200 mAh — HS 80	**80 km/h**	**410 g**	**~28 mnt**	**~38 km**	**~16 km**
3S 3000 mAh — HS 100	100 km/h	390 g	~18 mnt	~30 km	~12 km

Konfigurasi	Kecepatan	Kargo	Endurance	Range one-way	Radius RTH
4S 2200 mAh — HS 100	100 km/h	410 g	~18 mnt	~30 km	~12 km

8. Ringkasan Spesifikasi Final

Parameter	Nilai
Airfoil	**NACA 2412**
Semi-length	**1.52 ft (46.33 cm)**
Wingspan	**92.66 cm**
Root / Tip chord	**20.12 cm / 14.63 cm**
Taper ratio	0.7273
Sweep	0.0°
Wing area	0.1610 m²
Aspect ratio	5.3333
MAC	17.37 cm @ ±21.9 cm dari CL
Konfigurasi ekor	V-tail twin fin (45°)
Empty weight	**2.20 lb = 0.998 kg**
MTOW	**3.50 lb = 1.588 kg**
Payload	590 g
CG fwd	**0.62 ft = 4.95 cm dari LE = 28.5% MAC**
CG nominal ✓	**0.63 ft = 5.26 cm dari LE = 30.3% MAC**
CG aft limit	**0.65 ft = 5.87 cm dari LE = 33.8% MAC**
Wing loading	96.7 N/m²
Stall speed	**10.26 m/s (36.9 km/h)**
Cruise speed	**16.26 m/s (58.5 km/h)**
VNE	35.92 m/s
L/D max	11.49
Cruise power	36.3 W (0.049 HP)
Peak power	123 W (0.165 HP)
Endurance (cruise)	**~47 mnt**
Range one-way (cruise)	**~46 km**
Radius RTH (cruise)	**~19–21 km**
Cruise power HS 80	59.7 W (0.080 HP)
Endurance HS 80	**~28 mnt**
Range one-way HS 80	**~38 km**
Radius RTH HS 80	**~16 km**
Cruise power HS 100	93 W (0.125 HP)
Endurance HS 100	~18 mnt
Range one-way HS 100	~30 km

Parameter	Nilai
Radius RTH HS 100	~12 km
Motor	**2216, KV 1000 @ 3S**
Propeller (cruise)	APC 9×4.5
Propeller HS 80	APC 8×6
Propeller HS 100	**APC 9×7**
ESC	30 A BLHeli_32
Baterai	4S 2200 mAh / 3S 3000 mAh

UAV Satria Nano · NACA 2412 · Semi 1.52 ft · Root 0.66 ft · Tip 0.48 ft · Sweep 0° · V-tail
Empty 2.20 lb · MTOW 3.50 lb · Seluruh nilai performa adalah estimasi teoritis.